A Decision Layer for Perishable Supply Chains

What to buy, what to commit, what to promise, before supply is certain

Prepared for Sarah AlSaleh, General Partner, Outliers Venture Capital

THE DECISION GAP

Perishable operators commit supply to buyers before they know what their own suppliers will deliver. ERPs and inventory systems record what happened; they were not built to recommend what to commit next. That pre-commitment window has no standard decision layer, and McKinsey has found that most supply-chain planning still runs on spreadsheets. In perishables a wrong commitment resolves one of two ways, wasted product or a missed order, and the cost disappears into margin rather than onto a line anyone can manage.

WHY PERISHABLES, AND WHY THIS MATTERS FOR THE GULF

The product carries a clock, margins are thin, and the decision is irreversible once made. BCG's agrifood work identifies coordination failure, not only climate, as a primary driver of fresh-food supply-chain fragility. The Gulf feels this more sharply than most markets: the region imports the large majority of its fresh food, supply lines are long, and food security is a stated national priority in Saudi Arabia. Every imported pallet that spoils or order that goes short is a coordination failure paid for twice, once in margin and once in resilience. The operators who move that food have systems of record. They do not have decision support.

WHAT WE DO

We turn the messy CSVs an operator already exports into structured operational data, then run three things on top of it. A **diagnostic** that shows, in dollars, where margin is lost: waste, shortage, supplier fill and rejection, buyer fulfillment, and available-to-promise by product. A **compliance layer** that structures the lot-level traceability data regulators increasingly require (FSMA-204 in the US; the same structure serves Gulf import-traceability needs). And a **decision layer**, built on stochastic optimization over probabilistic supplier-yield and demand forecasts, that recommends how much to commit to each supplier and buyer and quantifies the overbuy and underbuy risk on each. We prove it by backtesting: we replay an operator's own history and measure what the recommendations would have saved, so trust comes from their numbers, not our claims.

WHERE WE ARE, STATED PLAINLY

The engine is built and gate-tested: demand and supplier-yield forecasters pass calibration gates, and the optimizer beats naive and heuristic policies across every configuration tested on a full-year simulated operator. All validation so far is on synthetic data; we are now recruiting the first pilot operators to run the backtest on real records. We have no live customers yet, and this letter is not a fundraise.

WHY YOU, SPECIFICALLY

Three reasons. You built data products for a living, at Dataminr's Middle East desk and as head of product at Wasphi, so you know better than most investors what separates a real decision product from a dashboard, and we want that scrutiny early. You know the Saudi and Gulf market, where the import-heavy fresh supply chain makes this decision more valuable than almost anywhere. And Outliers backs founders who build different, including from Purdue, where both of us trained as industrial engineers.

HOW A CONVERSATION COULD HELP BOTH SIDES

- **Your read on the product thesis:** is the pre-commitment decision the right wedge, and what would a Gulf operator need to see before trusting it?
- **A pointer, if it earns one:** an introduction to anyone in your network who touches agriculture or fresh food, whether an importer, distributor, grower, or operator, who would sit for a 30-minute discovery conversation. That would be worth more to us right now than capital.
- **For you:** an early, unfiltered window into a vertical-AI supply-chain team before there is anything to price, and a founding team you can watch execute from the first pilot onward.

We would value thirty minutes. Thank you sincerely for considering it.

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